MACHINE LEARNING · TIME SERIES · PERSONAL PROJECT

CryptoCast: Bitcoin Forecasting System

Forecasts Bitcoin prices 1, 3, and 7 days ahead, and compares a classic ML model with a deep learning model.

Built using a practice dataset · Kasthuri Sathish

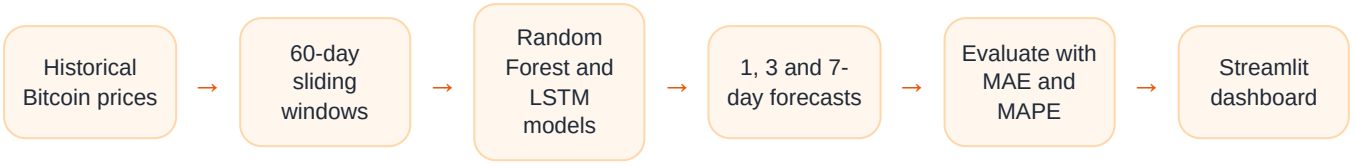
The problem

Bitcoin prices move quickly, and a single forecast hides how uncertain it is. People need forecasts for different time ranges, and a clear view of how much to trust each one.

What I built

A forecasting engine that predicts prices across 1, 3, and 7-day horizons, with an interactive Streamlit dashboard for scenario simulations, trend charts, and error metrics.

How it works



Key product decisions

- Offered three forecast horizons, because short and longer-term users need different answers.
- Compared Random Forest with an LSTM, to test whether deep learning was worth the extra complexity.
- Showed error metrics in the dashboard, so users can see how reliable each forecast is.
- Chose a 60-day lookback window to capture medium-term momentum without overfitting to daily noise.

Testing and learnings

- Random Forest had lower error on 1-day forecasts and trained faster, while the LSTM tracked 7-day trend direction better.
- Added momentum features, such as rolling moving averages and volatility spreads, to improve multi-day forecasts.
- Error grows with the forecast horizon, so 7-day forecasts are less reliable than 1-day forecasts.
- Limitation: forecasts are for learning, not financial advice.

Tech stack

Python · TensorFlow · Keras · Scikit-Learn · Time-Series Modeling · Streamlit